CATH PLAN · Case 2

74M, 78 kg · scheduled 09:30 · PCI to mid-LAD (NSTEMI) · pathway: known intervention

Synthetic teaching content. Not clinical advice. Not a medical device. Approved by the chair on the Switch card; still a planning draft, not an order.

Thresholds: PROPOSED values, not yet verified by the chair. Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is also unreviewed.

7	7.1	3 / 4	0
flags from 4 agents	contrast ratio (proposed cap 3.7)	angio findings flagged (calcium, flow, bifurcation, multivessel)	devices out of stock

Contrast ratio = planned contrast (mL) ÷ eGFR. The published ratios use creatinine clearance, not eGFR (docs/EVIDENCE.md E1, E2): risk rose from 2 (Gurm 2011) and 3.7 was a discriminating cutoff (Laskey 2007). No guideline sets a limit; Gurm: risk clearest at 3 or more, authors advise below 3 and preferably 2; Laskey 3.7 is a discriminating cutoff, not a threshold for harm. The chair confirms the basis. The packet flags this eGFR as unreliable while creatinine is changing.

Lesion map (Imaging)

Lesion	Vessel	Stenosis	TIMI	Calcium	Flag
L1	mid-LAD	90%	TIMI 2	calcium moderate	none
L2	mid-LCx	70%	TIMI 3	calcium none	none

Triggers: calcium moderate or severe; TIMI flow below 3; more than one lesion (multivessel). · Cath report: indication NSTEMI; access radial; hemodynamics not recorded

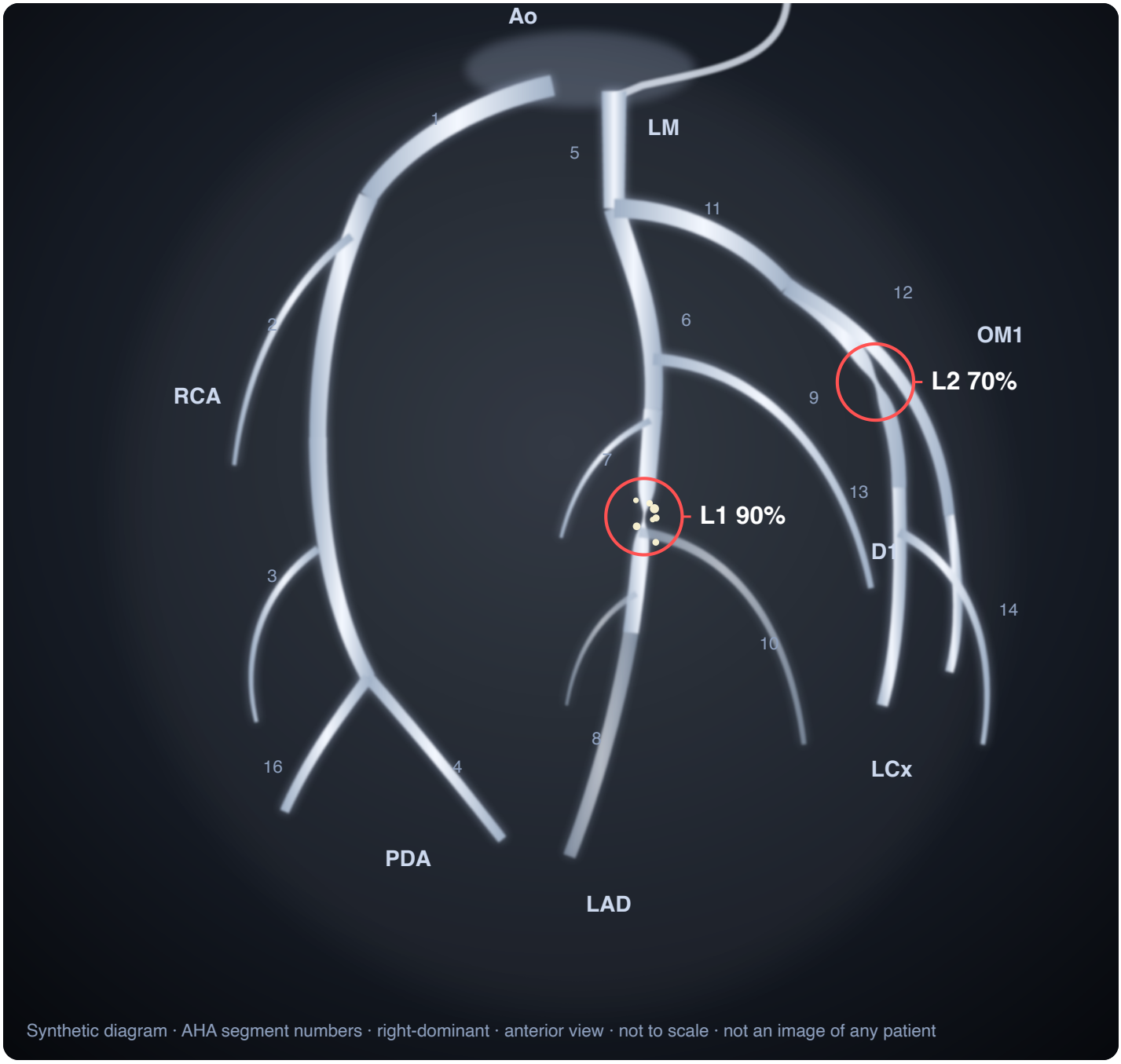
Rules that fired

- Contrast ratio above published cutoffs: 7.1 (170 mL ÷ eGFR 24) is above 3.7, a cutoff that discriminated early creatinine rise in Laskey 2007 (E2), and above 3, where risk was clearest in Gurm 2011 (E1). Observational cutoffs, not a limit; both used creatinine clearance, not eGFR. Chair to decide whether to stage.

If the chair stages (illustration from the packet, not a recommendation)

Stage	Lesions	Devices
Stage 1	L1 mid-LAD 90% (the procedure named in the packet)	7F guide, NC balloon 3.5 mm, DES 3.5x28 mm, standard PCI tray
Stage 2	L2 mid-LCx 70%	**not listed in the packet: Equipment to confirm**

Angiographic findings (synthetic)



L1 · mid-LAD (AHA segment 7)
90% stenosis · TIMI 2 (slow filling: the distal vessel is drawn fainter) · calcium moderate · flag none

L2 · mid-LCx (AHA segment 13)
70% stenosis · TIMI 3 (normal filling) · calcium none · flag none

Drawn from the angio lines in the packet by tools/angio.py: right-dominant tree, anterior view, AHA segment numbers, lumen narrowed to the stated percentage, calcium shown as bright specks, distal filling dimmed when TIMI flow is below 3. Solid red ring = stenosis of 70% or more; dashed amber ring = under 70%, or a CCTA estimate not yet confirmed. A Medina 1,1,1 flag is drawn as a narrowing of about half the proximal LAD and LCx. A teaching diagram, not an image of any patient and not to scale.

Equipment manifest

Item (from the packet)	Stock
7F guide	3 on hand
NC balloon 3.5 mm	3 on hand
DES 3.5x28 mm	2 on hand
standard PCI tray	3 on hand

Prep check (devices against findings)

Scope	Stage the devices named in the packet
Stock	all named devices in stock

On the shelf, by class (from the STOCK line)	
Guide catheters	7F guide 3, 6F guide 3
Diagnostic catheters	5F diagnostic catheters 6
Wires	support wire 3
Balloons	NC balloon 3.5 mm 3, NC balloon 3.0 mm 3
Stents (DES)	DES 4.0x24 mm 2, DES 3.5x28 mm 2, DES 3.0x18 mm 2
Intravascular imaging	IVUS catheter 0 (resupply ETA 13:00)
Support devices	Impella CP 2

Questions for the chair (the plan asks, it does not choose a device):

- 2 lesions on angiography and 1 stent named in the packet: chair to confirm what to stage.
- Calcium moderate on L1 mid-LAD (angiography): the packet names no calcium-modification device or intravascular imaging catheter. Chair to decide whether any is needed.
- No wire is named in the packet's device lines: chair to confirm.
- Calcium or left main on angiography, and IVUS catheter is 0 on hand with resupply ETA 13:00, after the scheduled 09:30 start: chair to decide how to proceed.

Flags from the agents

Imaging
- calcium moderate · L1 mid-LAD 90%
- TIMI below 3 · L1 mid-LAD 90% TIMI 2
- multivessel · L1 mid-LAD 90% · L2 mid-LCx 70%

Labs
- Cr 2.7 (2.2 yesterday) · Est. contrast 170 mL
- eGFR 24 · Est. contrast 170 mL

Meds
- heparin infusion running · radial access
- metformin on the list · Contrast given at the diagnostic angiogram yesterday

Equipment ✅ nothing flagged

Relay: Imaging → Equipment, Labs → Meds.

Summary and approvals

Ruling recorded	none yet
Status	✔ Chair ruling recorded: approved on the Switch card, 2026-11-04 20:48.
Flags	Imaging 3 · Labs 2 · Meds 2 · Equipment 0
Rules that fired	Contrast ratio above published cutoffs: 7.1 (170 mL ÷ eGFR 24) is above 3.7, a cutoff that discriminated early creatinine rise in Laskey 2007 (E2), and above 3, where risk was clearest in Gurm 2011 (E1). Observational cutoffs, not a limit; both used creatinine clearance, not eGFR. Chair to decide whether to stage.

For the chair to decide / check (the agents only flag):

- Thresholds in thresholds.json are proposed: the chair verifies them before this plan is used.
- Template logic (likelihood weights and bands, prep rules, Imaging triggers, Heart Team factors, stage assignment, Labs' eGFR cutoff) is unreviewed.
- Stage 2 devices are not listed in the packet: Equipment to confirm.

Approvals

Time	Request	Decision	By
20:48	this plan	Allow once	the chair, on the Switch card

Status

Approved by the chair on the Switch card.



Sources: packet (day.md) · briefs from Imaging, Labs, Meds, Equipment · thresholds.json · evidence table docs/EVIDENCE.md (E1 to E16, mostly secondary summaries, chair verifies). Generated 2026-11-04 20:48 by plan.py (template, no model).
Provenance: template a0c1d013d2 · thresholds 04864c04a6 · plan.py c9468ccb46 · packet id n/a. "ref:" on a flag names a published criterion; it is not an instruction.